

Summary

Experienced AI / ML Engineer with around 3+ years of experience, designing scalable machine learning solutions and distributed data pipelines to solve complex analytical and predictive modeling problems. Leveraging Python, PySpark, Apache Spark, deep learning, time-series forecasting, anomaly detection, and AWS-based cloud platforms to build and deploy production-grade ML systems using modern MLOps practices. Collaborating with cross-functional engineering, analytics, and compliance teams to deliver reliable, explainable, and high-impact data-driven solutions that improve operational efficiency and decision intelligence.

Technical Skills

Programming & Data Engineering: Python, SQL, PySpark, Scala, R, Bash, Data Structures, Algorithmic Modeling, Distributed Data Processing

Machine Learning & Statistical Modeling: Supervised Learning, Unsupervised Learning, Feature Engineering, Ensemble Modeling, Time-Series Forecasting, Anomaly Detection, Survival Analysis, Model Calibration, Imbalanced Data Handling

Deep Learning & Sequence Modeling: Neural Networks, Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Transformer Architectures, Sequence Modeling

LLMs & Generative AI Systems: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Embeddings, Vector Search, Prompt Engineering, Agentic Workflows, LoRA, QLoRA, Model Quantization

MLOps / LLMOps & Deployment: MLflow, Kubeflow, Docker, Kubernetes, KServe, TorchServe, FastAPI, CI/CD Pipelines, Feature Stores, Model Versioning, Automated Retraining, GPU Optimization

Monitoring, Explainability & Governance: SHAP, LIME, Model Explainability, Drift Detection, Data Quality Validation, Model Performance Monitoring, AI Observability, Responsible AI Practices

Cloud, Big Data & Distributed Systems: AWS (SageMaker, S3, EC2, EKS, EMR, Redshift), Azure Machine Learning, Google Cloud Platform, Apache Spark, Kafka, Flink, Snowflake, BigQuery, Hive

Databases, Analytics & Compliance: PostgreSQL, MongoDB, Cassandra, Redis, Elasticsearch, Data Warehousing, Analytical Modeling, Tableau, Power BI, Streamlit, Plotly, Data Governance, Model Risk Management, Auditability, Data Lineage, Access Control, GDPR Compliance, SOX Compliance, AI Ethics Frameworks

Experience

Northern Trust, USA AI / ML Engineer - Engineered distributed feature engineering pipelines using Python, PySpark, SQL, Apache Spark, AWS EMR, Amazon S3, and Snowflake to analyze large financial datasets and generate predictive features for portfolio risk forecasting models. - Developed time-series forecasting and ensemble models using LSTM, Transformer architectures, supervised learning, PyTorch, TensorFlow, and feature engineering, improving portfolio volatility prediction accuracy by 28% across institutional investment portfolios. - Implemented anomaly detection systems using Isolation Forest, Autoencoders, unsupervised learning, Apache Spark, and Kafka streaming, identifying abnormal trading activity and liquidity disruptions across multi-asset investment portfolios. - Architected financial intelligence workflows using Large Language Models, Retrieval-Augmented Generation, embeddings, vector search with Elasticsearch, prompt engineering, LoRA, and QLoRA, enabling automated analysis of regulatory documents and investment reports. - Optimized machine learning deployment pipelines using MLflow, Kubeflow, Docker, Kubernetes, KServe, CI/CD pipelines, and AWS EKS, reducing manual portfolio risk monitoring workload by 40% through scalable AI infrastructure. - Built explainable AI and compliance analytics frameworks using SHAP, LIME, Tableau, Streamlit, PostgreSQL, and data governance frameworks, increasing regulatory risk detection efficiency by 35% across automated financial document intelligence systems.	Jul 2025 – Present
Cybage Software, India Machine Learning Engineer - Designed end-to-end machine learning pipelines using Python, SQL, Apache Spark, and AWS, improving transaction success and compliance risk detection accuracy by 34% across large-scale fintech datasets. - Developed predictive classification models using Logistic Regression, Random Forest, XGBoost, and Gradient Boosting, reducing payment failures and AML false positives by 41% through structured feature engineering. - Implemented behavioral segmentation and anomaly detection using KMeans, DBSCAN, and Isolation Forest, improving identification of abnormal transaction patterns by 29% in digital payment systems. - Built sequential transaction models using LSTM, time-series feature extraction, and deep learning workflows, enhancing detection of retry and layering patterns by 26% across transactional data streams. - Optimized retry-success prediction services using machine learning scoring APIs, batch processing, and Docker-based deployments, reducing unnecessary retries by 37% and stabilizing payment operations. - Improved compliance investigation efficiency by 32% by integrating SHAP explainability, model performance dashboards, and risk scoring outputs, enabling faster and more accurate manual reviews. - Stabilized production ML workflows using Airflow orchestration, MLflow experiment tracking, and scheduled retraining, achieving 45% higher model reliability across fintech and banking clients.	Jan 2021 – July 2023

Education

Saint Louis University, USA Master of Science in Artificial Intelligence	May 2025
St. Martin's Engineering College, India Bachelor of Technology in Computer Science and Engineering	Jul 2023

Projects

AI News Summarizer Bot - Collected and processed live news data using RSS feeds, performed text cleaning and transformation, and applied NLP techniques to generate concise summaries for analytical consumption. - Built an end-to-end data pipeline using Python and a pretrained transformer model to analyze content trends and enable real-time, category-based insights.
